# AI Health Report & Medicine Analyzer

*Project Proposal*

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## 1. Project Overview

### Project Topic

Most people get their blood test results and have no idea what they mean. The numbers are there, the reference ranges are there, but nothing explains whether you should actually be concerned or not. We want to build a web app that takes a lab report PDF, reads it, and explains the results in plain language using AI.

There is also a medicine checker where users enter the medicines they are taking and the app flags any combinations that are known to be risky.

### Objective

- Let users upload a lab report PDF and get a readable explanation of their results.
- Clearly show which values are Normal, High, or Low.
- Use the Gemini API to explain what the results mean and what to ask a doctor.
- Include a medicine interaction checker.
- Deployment on Streamlit Cloud.

## 2. System Description

### Problem Background

Patients leave labs with printed reports they cannot interpret. Even when a value is flagged as high, there is no explanation of what that means or how urgent it is. People either Google it and panic or just ignore it entirely.

The same problem exists with medicines. Many people take multiple medications without knowing whether they interact. There is no simple, accessible tool a regular person can use to check this.

### Innovations Introduced

- PDF upload and automatic extraction of lab values — no manual typing needed.
- Color-coded results table showing Normal, High, or Low for each value.
- AI-generated explanation of results written for someone with no medical background.
- Medicine interaction checker that flags unsafe combinations.

## 3. AI Approach and Methodology

### LLM — Gemini API

We are using the Google Gemini API to generate explanations of lab results and answer follow-up questions. The prompts are written so the model explains values simply, avoids making diagnoses, and always tells the user to see a doctor if something looks off.

For the medicine checker, the exact implementation will be decided during development — either Gemini alone or combined with an external drug database.

### Classification Logic

Before the AI is involved, each value is compared against standard reference ranges and tagged Normal, High, or Low. This is a straightforward lookup so the table is always accurate regardless of the model output.

### Other Techniques

- PDF text extraction from uploaded lab reports.
- Text parsing to identify parameter names and values.
- Keep AI responses safe and relevant.

## 4. System Rules and Mechanics

### Workflow

User Input → PDF Parsing → Value Classification → AI Analysis → Output

### Process Steps

1. User uploads a lab report PDF or enters values manually.
2. System extracts text and identifies test names and values.
3. Each value is classified as Normal, High, or Low.
4. Structured data is sent to Gemini which writes a plain-language summary.

5. Results and explanation are displayed together.
6. User can enter medicine names to check for interactions.

**Success Conditions**

- Values are extracted and classified correctly.
- The AI explanation is clear to someone with no medical background.
- Medicine checker returns a useful response.
- Live and accessible from a public link.

## 5. Libraries and Tools

| Tool | Purpose |
|---|---|
| **Streamlit** | Web UI  builds the full interface without HTML or JavaScript |
| **Google Gemini API** | AI model for generating explanations |
| **pdfplumber** | Extracts text from uploaded PDF lab reports |
| **Pandas** | Structures the extracted values into a table |

## 6. Implementation Plan and Timeline

| Week | Focus | Plan |
|---|---|---|
| **Week 1** | **Setup** | Set up environment, install dependencies, get API key, build basic Streamlit layout. |
| **Week 2** | **Report Reading** | PDF upload, text extraction, value classification, results table display. |
| **Week 3** | **AI** | Connect Gemini API, write prompts, show explanations with results. |
| **Week 4** | **Medicine Check** | Build medicine input section and get interaction checker working. |
| **Week 5** | **Wrap Up** | Test with real reports, fix bugs, add disclaimer, deploy. |

## 7. References


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